MoleculeLens: Interpretable Drug–Text Retrieval via Salient Molecular Substructures

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Abstract

We present *MoleculeLens*, a framework for interpretable drug–text retrieval that makes every alignment decision traceable to molecular substructures. The model combines frozen ECFP4 fingerprints and a frozen biomedical sentence encoder with learned linear projection heads in a shared embedding space. Because the molecule-side map is linear, the model admits a closed-form linear saliency approximation over ECFP4 bits; on held-out scaffold pairs, this proxy tracks the exact normalized-score gradient strongly (mean Pearson 0.905) and the exact leave-one-bit-out score even more closely (mean Pearson 0.947). We evaluate the framework on a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs under a Bemis–Murcko scaffold-disjoint split designed to reduce memorization and expose lexical leakage. On this harder setting, MoleculeLens remains competitive with retrieval baselines while enabling analyses that black-box models cannot: pharmacophore-level explanations for correct matches, attribution fidelity diagnostics for failed retrievals, a same-model substructure-level account of drug-name leakage, and a descriptive layer-wise view of where drug-retrievable information emerges in the frozen text encoder. These results position MoleculeLens as an auditable approach to studying what molecular features drive drug–text alignment under realistic scaffold shift.

1 Introduction

Drug–text alignment asks whether molecular structures and mechanism descriptions can be embedded in a common space for retrieval. A useful system should generalize beyond memorized scaffolds and should reveal which molecular features support each match. Current evaluations make both questions difficult: drug names, target labels, and scaffold-correlated training examples can inflate retrieval scores, while standard dual encoders provide little evidence about the substructures behind a prediction.

MoleculeLens addresses these issues with a deliberately constrained architecture. It combines frozen ECFP4 fingerprints and a frozen biomedical text encoder with learned linear projection heads. The linear molecule-side bridge yields a closed-form saliency proxy over ECFP4 bits, which can be decoded with RDKit into atom-level substructures. We validate this proxy against exact gradients and exact leave-one-bit-out score drops, then use it to audit retrieval decisions and drug-name leakage under a scaffold-disjoint ChEMBL split.

This paper makes four contributions. First, we introduce validated closed-form linear saliency for drug–text retrieval. Second, we construct a reproducible scaffold-disjoint benchmark of 2,699 approved-drug mechanism pairs. Third, we show that leakage from explicit drug names is visible not only in retrieval scores but also in which molecular substructures the fixed rich model treats as salient. Fourth, we provide a descriptive contrastive logit-lens diagnostic showing that drug-retrievable CLS information in the frozen text encoder appears almost entirely at the final layer.

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2 Related Work

Molecule–text retrieval and molecular representation learning. Drug–text and molecule–language models include retrieval-oriented systems such as Text2Mol [Edwards et al., 2021], molecular language models such as MolT5 [Edwards et al., 2022] and BioT5 [Pei et al., 2023], graph-language systems such as MolCA [Liu et al., 2023], and multimodal biomedical pretraining such as KV-PLM [Zeng et al., 2022]. Graphormer [Ying et al., 2021] and Uni-Mol [Zhou et al., 2023] are important molecular representation baselines, though not direct text-query retrieval systems without additional alignment. MoleculeLens is complementary: it emphasizes scaffold-disjoint drug–mechanism retrieval and auditable substructure-level explanations rather than maximum-capacity generation.

Leakage, scaffold splits, and interpretability. Scaffold splits are a standard guardrail against overoptimistic molecular generalization estimates [Bemis and Murcko, 1996]. We combine this with explicit drug-name and mechanism-only ablations because identity-bearing text fields can become shortcuts. The interpretability component is motivated by work on linear feature analysis and attribution graphs [Bricken et al., 2023, Templeton et al., 2024, Lindsey et al., 2025], but the features here are not discovered post hoc: ECFP4 bits already correspond to labelled circular molecular environments [Rogers and Hahn, 2010] recoverable with RDKit [RDKit contributors, 2023].

3 Benchmark and Method

Dataset and task. We construct the benchmark from ChEMBL [Zdrazil et al., 2024] by filtering for approved drugs (`max_phase=4`), valid mechanism text, and parseable canonical SMILES. The final dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds. The primary text field concatenates mechanism of action, target name, action type, and drug name:

$$\texttt{text_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{1}$$

The main split uses seed 0 over unique scaffolds: 2,019 train examples, 245 validation examples, and 435 held-out test examples whose scaffolds are unseen during training. We report text-query retrieval against a molecule gallery using single-label diagonal Recall@1, MRR, Recall@5, and Recall@10 for apples-to-apples comparison with baselines. Appendix A gives additional evaluation details, including the parent-structure handling used for attribution analyses.

Model. For drug i, RDKit computes a radius-2, 2048-bit ECFP4 fingerprint x_i [Rogers and Hahn, 2010]. For text i, a frozen biomedical sentence encoder produces a CLS-style embedding z_i. MoleculeLens learns only two linear projection heads:

$$b_i^{(m)} = W_m x_i + b_m, \qquad b_i^{(t)} = W_t z_i + b_t, \tag{2}$$

followed by L_2 normalization and cosine scoring. Training uses symmetric InfoNCE with same-target weighting and a hardest-same-target margin term. The scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 epochs. All encoders are frozen; only W_m, b_m, W_t, b_t are trained. Architecture and implementation details are in Appendix B.

4 Retrieval Results

Table 1 shows the main comparison. MoleculeLens has the strongest scaffold-split Recall@1 and MRR among the evaluated cross-modal baselines, improving Recall@1 over MolPrompt by 9.6 points and over KV-PLM by 17.0 points. MolPrompt has higher Recall@5/10, so the claim is not leaderboard dominance: MoleculeLens is strongest at top-rank retrieval while remaining competitive at broader candidate coverage. Full-gallery rows are included only as context; the Thin Bridges global result is a memorization-heavy reference, not the main scientific comparison.

Text-field and leakage controls. Because `text_rich` includes target/action metadata and an explicit drug field, we report three complementary controls in Table 2. Removing only the drug field and retraining gives R@1 = 0.110, a 51.0% relative drop. Holding the rich model fixed and removing only `Drug: X.` at inference gives R@1 = 0.067, a stricter 70.4% same-model causal diagnostic.

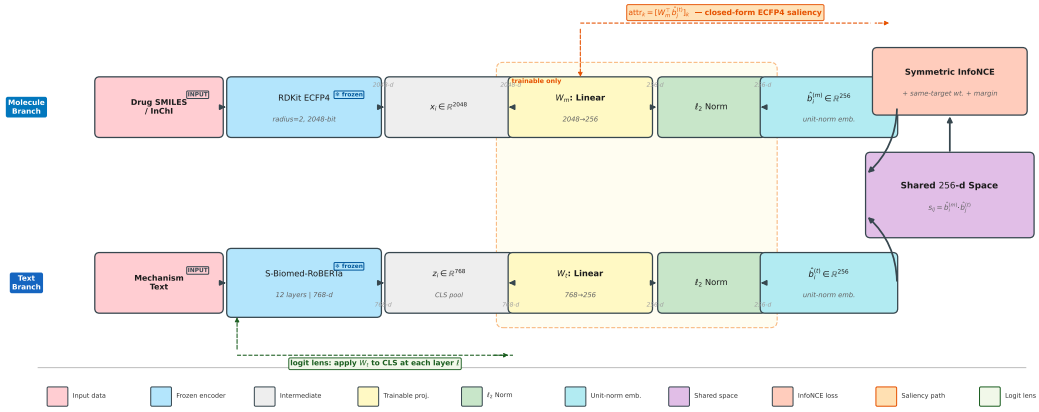

Figure 1: **MoleculeLens architecture.** Frozen ECFP4 and biomedical text encoders are connected by learned linear projection heads. The molecule-side linear map enables direct ECFP4-bit saliency; the text-side projection is reused for the layer-wise logit-lens diagnostic.

Table 1: Primary retrieval comparison. All rows use single-label diagonal evaluation. Full-gallery rows are context only; scaffold rows are the main comparison.

Model	Eval set	R@1	MRR	R@5	R@10
MoleculeSTM (zero-shot)	Full (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold (N=435)	0.002	0.015	0.011	0.025
MoleculeLens	Scaffold (N=435)	**0.225**	**0.304**	0.384	0.462
MolPrompt	Scaffold (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold (N=435)	0.055	0.121	0.172	0.251

Training on mechanism text alone, removing target name, action type, and explicit drug name, gives R@1 = 0.124 and MRR = 0.226. Thus target/action fields contribute real signal, but mechanism-only retrieval remains nontrivial and close to MolPrompt scaffold R@1.

The non-overlapping Recall@1 intervals for MoleculeLens `text_rich` and MolPrompt support the reliability of the top-rank improvement despite the 435-example test set. Five additional scaffold splits with a simplified proxy objective give R@1 0.213 ± 0.045, and a five-seed full-loss spot check using the same seeds 0–4 gives R@1 0.277 ± 0.081; full per-seed tables and the robustness figure are in Appendix C. The three leakage numbers answer different questions: the 51.0% retrained drop is the fairest model-comparison number, the 70.4% same-model drop is the causal diagnostic for the fixed rich model, and the 22.9% multi-seed proxy mean measures partition sensitivity rather than canonical leakage.

5 Substructure-Level Saliency

The implemented score is

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \text{normalize}(W_m x_i + b_m) \cdot \text{normalize}(W_t z_j + b_t). \tag{3}$$

Because W_m is linear, we use the closed-form ECFP4 saliency proxy

$$\text{attr}_k(i,j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{4}$$

where $\hat{b}_j^{(t)} = \text{normalize}(W_t z_j + b_t)$. Equation 4 is a fast ranking proxy rather than an exact causal decomposition: the exact normalized score also depends on the molecule-side bias and $\|W_m x_i + b_m\|$. We therefore validate it directly against exact gradients and exact leave-one-present-bit-out score drops over all 435 scaffold test pairs.

Table 2: Key controls on the canonical scaffold split. Confidence intervals are 95% intervals for Recall@1; MoleculeLens uses exact bootstrap resampling and other baselines use Wilson intervals.

Condition	R@1	MRR	R@1 95% CI
MoleculeLens `text_rich`	0.225	0.304	[0.186, 0.267]
MoleculeLens retrained no-drug	0.110	–	[0.083, 0.140]
MoleculeLens same-model no-name	0.067	–	[0.044, 0.092]
MoleculeLens mechanism-only	0.124	0.226	–
MolPrompt scaffold	0.129	0.256	[0.101, 0.164]
KV-PLM scaffold	0.055	0.121	[0.037, 0.081]

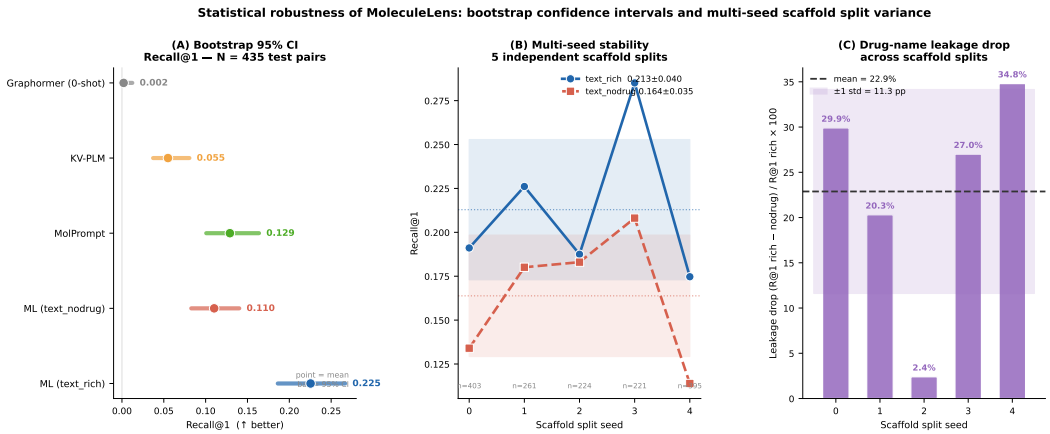

Figure 2: **Statistical robustness summary.** Bootstrap intervals, multi-seed scaffold variance, and per-seed drug-name leakage drops show that the main Recall@1 gap is statistically reliable while leakage varies with scaffold partition.

Table 3 supports rank-based use of the saliency proxy. It tracks the exact normalized-score gradient strongly and the exact bit-drop score even more closely. The identical-text test addresses a key conceptual concern: the exact top-10 bit-drop sets are almost never identical across distinct drugs sharing the same no-drug query, so the implemented normalized score is strongly drug-specific. The proxy also remains faithful for failed retrievals, which means incorrect predictions remain auditable decisions of the learned scoring function rather than cases where the attribution method collapses. Appendix D provides additional case studies and aggregate bit statistics.

Same-model leakage attribution. We next ask whether the same rich model uses the same substructures when the explicit drug name is removed at inference. For each test pair, let $\mathcal{T}_i^{\text{rich}}$ and $\mathcal{T}_i^{\text{nodrug}}$ be the top-10 saliency-bit sets under the original query and the same query with `Drug: X.` removed. We compute

$$J_i = \frac{|\mathcal{T}_i^{\text{rich}} \cap \mathcal{T}_i^{\text{nodrug}}|}{|\mathcal{T}_i^{\text{rich}} \cup \mathcal{T}_i^{\text{nodrug}}|}. \tag{5}$$

Mean overlap is only 0.356 ± 0.161; 82/435 pairs have $J_i < 0.2$, 36/435 have $J_i > 0.6$, and 13/435 are leakage candidates that are correct under `text_rich`, wrong under the same-model no-name query, and have $J_i < 0.2$. This converts the score-level leakage drop into a structural claim: for a substantial subset of test pairs, removing the name changes which molecular features drive retrieval even though model weights are fixed.

6 Contrastive Logit Lens

We adapt the logit-lens idea [nostalgebraist, 2020] by applying the trained text projection head W_t to the CLS state at each S-Biomed-RoBERTa layer. The diagnostic exactly replays the main rich model at layer 12: the reconstructed embeddings match the saved outputs within 1.21×10^{-3} maximum absolute error and recover R@1 = 0.225. Layers 0–11 are near chance for both rich and retrained

Table 3: Validation of Equation 4. All statistics are over present ECFP4 bits. The final rows test drug-specificity using distinct-drug pairs with identical no-drug text.

Metric	Value
Mean Pearson(attr, exact gradient)	0.905
Mean Pearson(attr, exact bit-drop)	0.947
Mean top-10 Jaccard(attr, exact bit-drop)	0.710
Incorrect Rank-1 pairs: Pearson(attr, bit-drop)	0.970
Incorrect Rank-1 pairs: top-10 Jaccard	0.761
Identical no-drug text: mean exact top-10 Jaccard	0.136
Identical no-drug text: non-identical exact top-10 sets	952/954

no-drug text (R@1 between 0.002 and 0.011), while layer 12 jumps to R@1 = 0.225 for `text_rich` and 0.113 for `text_nodrug`. This is best viewed as a descriptive sanity check: retrieval signal and the rich-vs-no-name gap both appear in the final contextualized CLS representation, not gradually across intermediate layers. The full layer table, family breakdown, and attention visualizations are in Appendix E.

7 Discussion and Limitations

MoleculeLens is not a claim that frozen ECFP4 plus a linear bridge solves scaffold-generalized drug–text retrieval. Its narrower contribution is that a lightweight model remains competitive at top-rank scaffold retrieval while making its decisions inspectable at the ECFP4-substructure level. The saliency proxy is validated against exact score-derived quantities, remains faithful on retrieval failures, and exposes same-model drug-name leakage as a shift in salient molecular features.

The benchmark has important limitations. The rich text field includes target and action metadata, and the mechanism-only ablation confirms that these fields contribute retrieval signal; the rich-text result should therefore be interpreted as curated mechanism-metadata retrieval rather than pure free-text mechanism understanding. The dataset covers FDA-approved drugs with curated ChEMBL mechanisms and does not establish generalization to investigational compounds or non-ChEMBL prose. Finally, the baseline set does not include adapted MolT5, BioT5, Uni-Mol, MolCA, or chemistry-LLM retrieval systems; these require nontrivial task adaptation and remain important future comparisons.

8 Conclusion

On a 2,699-pair ChEMBL benchmark with scaffold-disjoint evaluation, MoleculeLens outperforms MolPrompt and KV-PLM on Recall@1 and MRR while clearly exposing the role of lexical and target/action metadata. Its central value is interpretability: the linear molecular bridge yields a validated, efficient ECFP4 saliency lens that can audit correct matches, failed retrievals, and same-model leakage. This makes MoleculeLens a useful foundation for future work on stronger encoders, text-identifiability audits, and feature-level analysis of drug–text alignment.

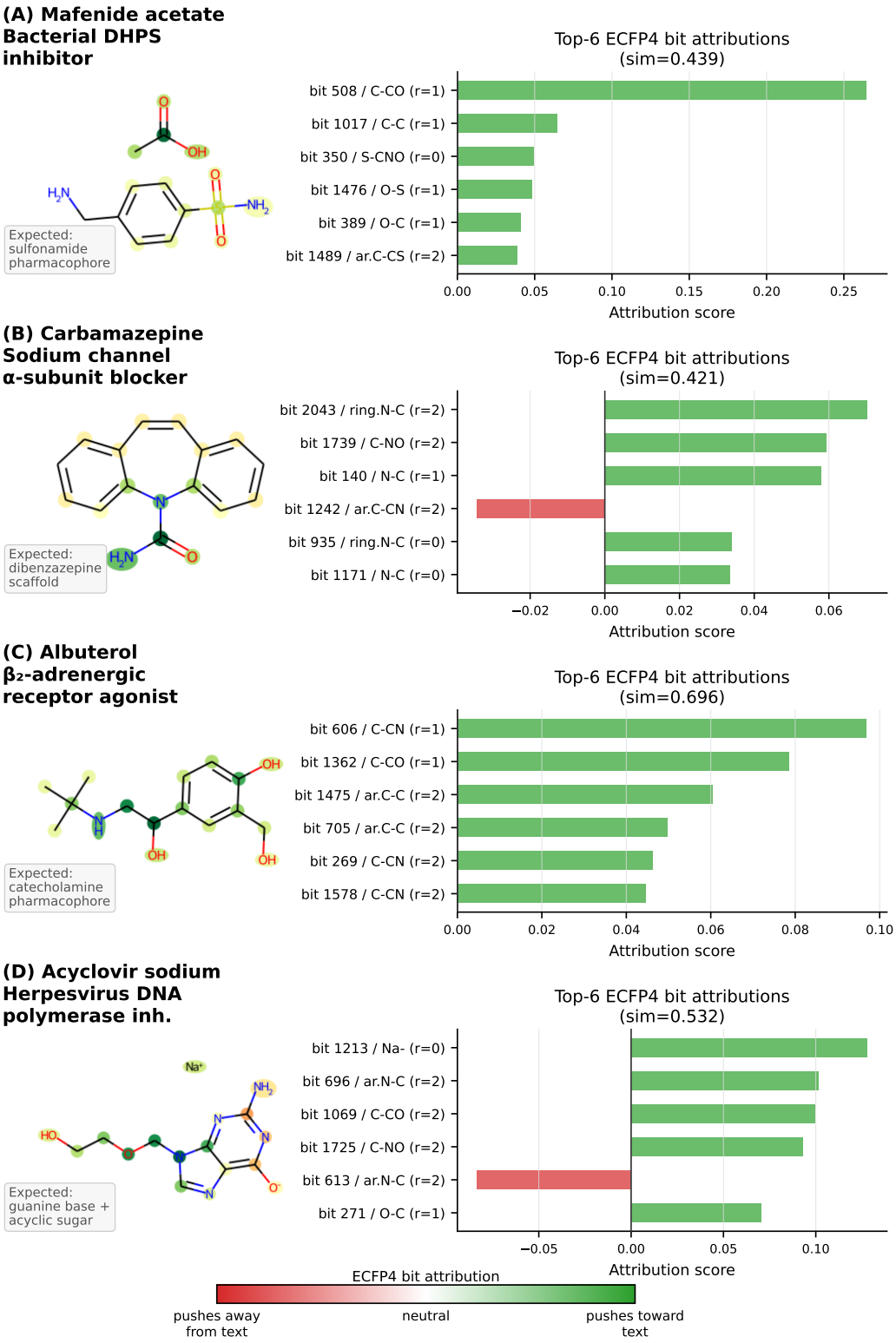

Figure 3: **ECFP4 linear saliency for four correctly retrieved pairs.** Each panel has three columns. *Left:* molecule with atoms colour-coded by summed attribution (green pushes toward the text match; red pushes away). *Centre:* the ECFP4 circular environment for each of the top-6 bits; the blue atom is the center and light-blue atoms form its radius-r neighborhood. *Right:* attribution scores; axis labels decode each bit as `[ar.|ring.]element-neighbors(r=radius)`, where `ar.` = aromatic, `ring.` = non-aromatic ring, and radius is the circular neighborhood depth (0–2 bonds). High-attribution substructures align with the expected pharmacophoric regions annotated in each panel.

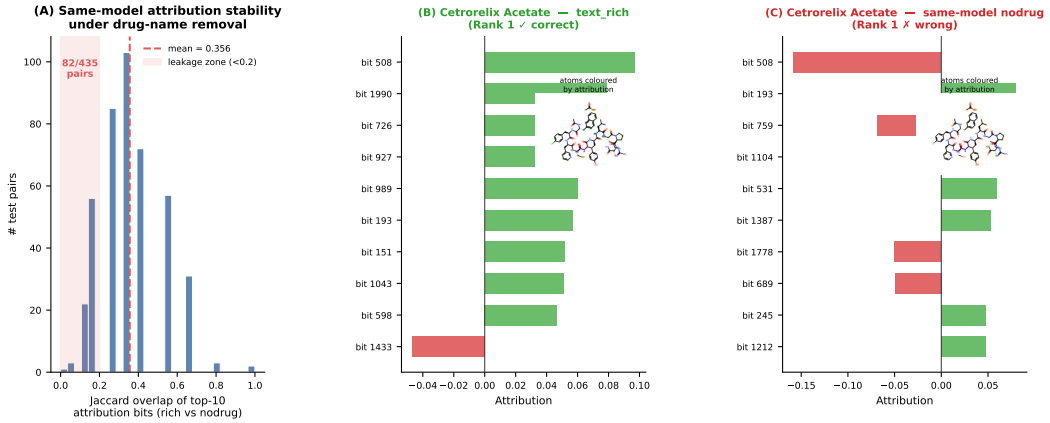

Figure 4: **Same-model drug-name ablation shifts salient substructures.** Removing only the explicit drug-name span from the query changes top-saliency ECFP4 bits for many test pairs; Cetrorelix acetate illustrates a low-overlap leakage candidate.

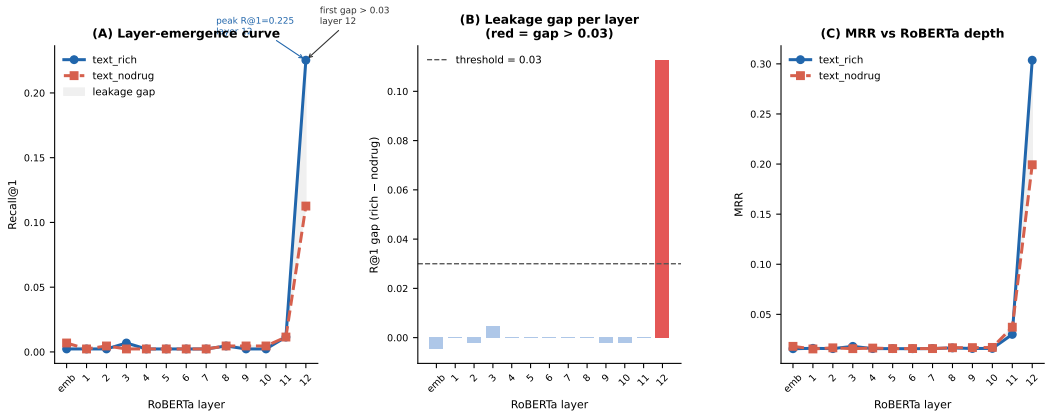

Figure 5: **Contrastive logit lens across RoBERTa layers.** Retrieval is near chance until the final transformer layer, where both rich retrieval signal and the no-name gap emerge.

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A Additional Benchmark and Evaluation Details

The primary retrieval direction is text query to molecule gallery because this is the direction used by the attribution and logit-lens analyses. All main-table rows use single-label diagonal evaluation: a prediction is counted correct only when the exact matching row is returned. This is intentionally conservative for drugs with multiple ChEMBL mechanism rows, but it is the only consistent apples-to-apples protocol for baselines where per-example ranked lists are not available.

Several test drugs appear under multiple target–action rows. For attribution analyses, same-drug and wrong-but-close filtering use RDKit `FragmentParent` standardisation over SMILES to avoid treating salt or formulation variants as independent structures. In the canonical scaffold test set, 193 of 435 rows belong to 76 parent structures with multiple mechanism rows.

B Architecture and Implementation Details

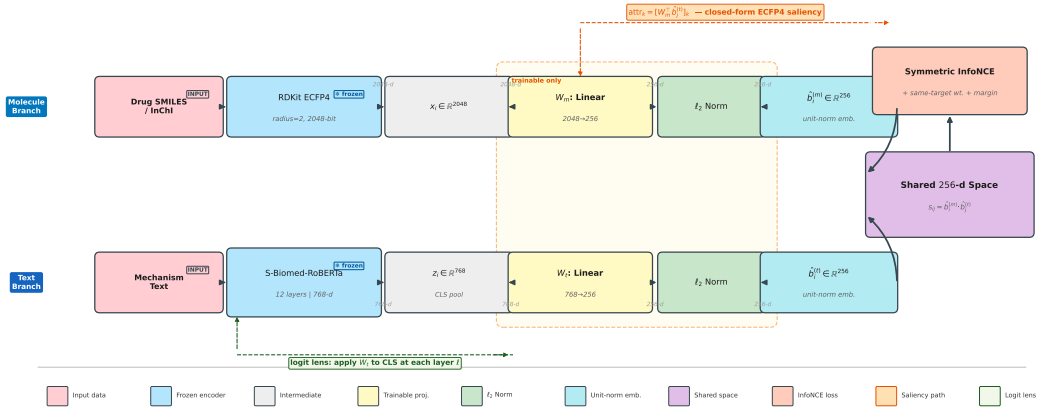

Figure 6: **MoleculeLens architecture.** A fixed ECFP4 molecule representation and a frozen biomedical text encoder are aligned by learned linear projection heads in a 256-dimensional shared space. Only the linear heads are trained; the encoders remain frozen. The molecule-side linear map enables the ECFP4 saliency proxy used in the main text, while the text-side projection head is reused for the contrastive logit-lens diagnostic.

The model trains for 100 epochs with batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, and margin 0.1. Training the projection heads takes under 30 seconds per run on a single NVIDIA GPU. ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes.

C Robustness Details

Table 4: Multi-seed scaffold split variance for MoleculeLens using the simplified proxy objective. Five independent Bemis–Murcko splits are used; only linear projection heads are retrained per seed. Leakage drop is $(R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}}$.

Seed	n_{test}	R@1	R@5	R@10	MRR	Leakage drop
0	403	0.191	0.377	0.459	0.287	29.9%
1	261	0.226	0.479	0.544	0.342	20.3%
2	224	0.188	0.393	0.496	0.296	2.4%
3	221	0.285	0.525	0.634	0.402	27.0%
4	395	0.175	0.352	0.446	0.269	34.8%
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9%**
Std	–	0.045	0.070	0.073	0.053	12.6%

Table 5: Five-seed full-loss check. These runs use the same same-target weighting and hardest-same-target margin term as the main MoleculeLens model. Seeds 0–4 were selected as the same consecutive seed set used by the proxy analysis before evaluating full-loss results.

Seed	n_{test}	R@1	R@5	R@10	MRR
0	403	0.221	0.335	0.424	0.291
1	261	0.295	0.479	0.533	0.381
2	224	0.259	0.375	0.442	0.328
3	221	0.407	0.561	0.629	0.484
4	395	0.203	0.342	0.451	0.284
Mean	–	**0.277**	**0.418**	**0.496**	**0.354**
Std	–	0.081	0.098	0.085	0.083

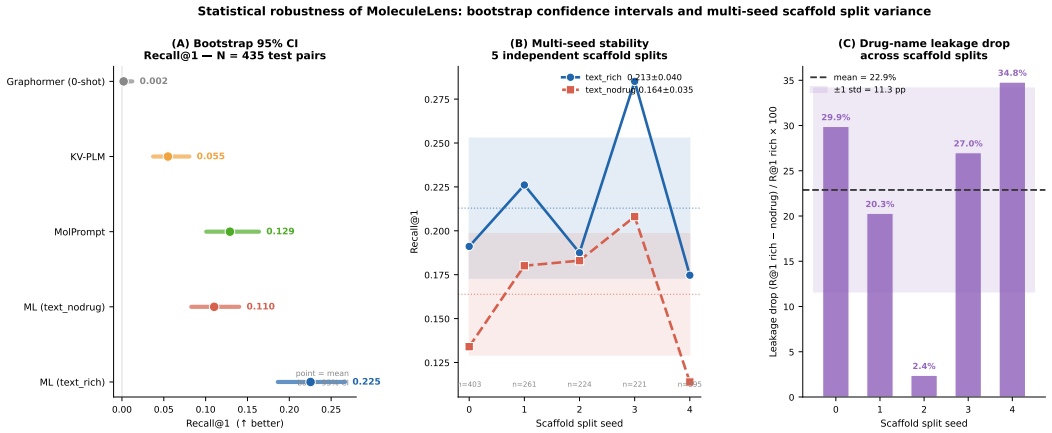

Figure 7: **Statistical robustness of MoleculeLens.** Bootstrap intervals, multi-seed scaffold variance, and per-seed drug-name leakage drops.

D Additional Attribution Analyses

For each test pair, we compute Equation 4, restrict to ECFP4 bits present in the molecule, rank bits by absolute saliency, and decode top-ranked bits with RDKit. Among eight correctly retrieved examples spanning antibacterial, antiviral, nuclear-receptor, opioid, sulfonamide, and diagnostic-agent mechanisms, the highest-saliency bits correspond to pharmacophoric or mechanism-defining regions such as steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

Table 6: Most frequent high-saliency ECFP4 bits across 435 test pairs. Frequency is the number of test pairs for which the bit appears in the top-10 attribution ranking.

Bit index	Frequency	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

Bit 1538 is the most frequent high-saliency feature, appearing in the top-10 for 85 pairs. RDKit decoding maps it to a benzene-ring environment with at least one heteroatom substituent at radius 2,

a common pharmacophore anchor across several major test-set families. The frequency drops sharply after the top few bits, indicating that no small universal shortcut dominates the saliency landscape.

E Contrastive Logit-Lens Details

Table 7: Recall@1 and MRR at key RoBERTa layers under both text conditions. Layers 0–11 carry essentially no drug-retrievable signal; meaningful retrieval appears at layer 12.

Layer	text_rich		text_nodrug		Gap
	R@1	MRR	R@1	MRR	R@1
0	0.002	0.016	0.007	0.018	–0.005
3	0.007	0.018	0.002	0.016	0.005
6	0.002	0.016	0.002	0.016	0.000
9	0.002	0.016	0.005	0.017	–0.002
11	0.011	0.030	0.011	0.037	0.000
12	**0.225**	**0.304**	**0.113**	**0.199**	**0.113**

Table 8: Peak Recall@1 at layer 12 by mechanism family under `text_rich`.

Mechanism family	n	R@1
Nuclear Receptor agonist/antagonist	18	0.611
Kinase inhibitor	11	0.545
Antiviral (DNA polymerase inhibitor)	15	0.533
Antibacterial	25	0.440
Diagnostic agent	15	0.333
GPCR agonist/antagonist	129	0.271
COX Inhibitor	16	0.250
Ion Channel blocker	41	0.195
Other	165	0.152

F Limitations and Future Work

Dataset scope. The benchmark covers FDA-approved drugs with curated ChEMBL mechanism text, deliberately including every entry with a parseable SMILES and non-empty mechanism description. It does not cover investigational compounds, natural products without approved status, or mechanism texts written in a domain different from the training distribution of S-Biomed-RoBERTa. Generalisation to these settings is an open question.

Lexical leakage. Drug-name leakage is measured by removing the explicit `Drug: X.` field. Implicit lexical cues (e.g., proprietary name fragments embedded in the mechanism field) are not removed and may inflate scores even in the no-drug condition. The target name and action type fields in `text_rich` introduce an additional partial target-identity lookup signal beyond the explicit drug-name field; the mechanism-only ablation quantifies this effect but does not eliminate all possible lexical clustering.

Baseline scope. The primary comparison is restricted to cross-modal methods that score text queries against molecular representations. Same-modality text lookup controls are not included as competing baselines because they access gallery text rather than structure. Future benchmark releases should include a separate text-identifiability audit, but those diagnostics should be reported separately from cross-modal retrieval baselines.

SAE and cross-modal crosscoder. The shared 256-dimensional embedding space is amenable to sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] to measure whether alignment dimensions are monosemantic or polysemantic. A cross-modal crosscoder [Lindsey et al., 2025] trained on concatenated drug–text embedding pairs would further distinguish cross-modal alignment features from text-only leakage features. These analyses are left for future work.

G Broader Impacts

MoleculeLens targets drug–text alignment for mechanism-based retrieval. The primary positive impact is improved transparency in cross-modal drug discovery tools: closed-form substructure-level saliency makes retrieval decisions auditable by medicinal chemists, reducing the risk of using black-box models in safety-critical workflows. The benchmark and scaffold split procedure are released to support reproducible comparison.

We do not foresee significant negative societal impacts. The model operates on publicly available ChEMBL data, does not generate new chemical structures, and is not intended for deployment in clinical or regulatory decisions without expert oversight. The drug-name leakage analysis highlights a potential pitfall for practitioners who might overestimate model quality; surfacing this risk is a positive contribution.

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   Justification: Section 3.3 specifies the Bemis–Murcko scaffold split sizes (2,019 train / 245 val / 435 test), and Section 5.2 lists all training hyperparameters. The Adam optimizer is used by default; the text encoder is frozen throughout.

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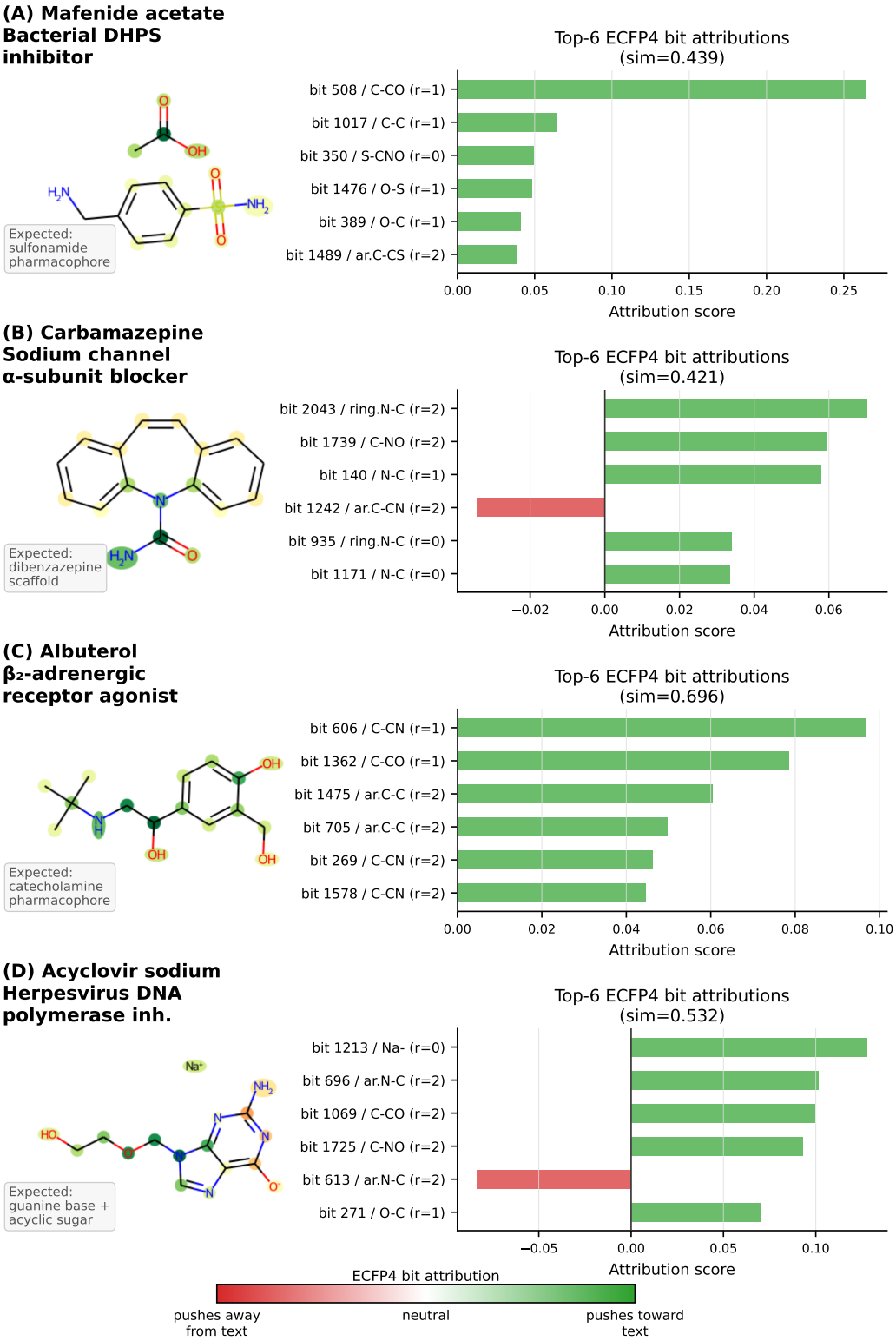

Figure 8: **ECFP4 linear saliency examples for correctly retrieved pairs.** Same layout as Figure 3: atom-coloured molecule, ECFP4 circular environment thumbnails (blue = center atom), and decoded substructure attributions. Axis labels use the format `[ar.|ring.]element-neighbors(r=radius)`.

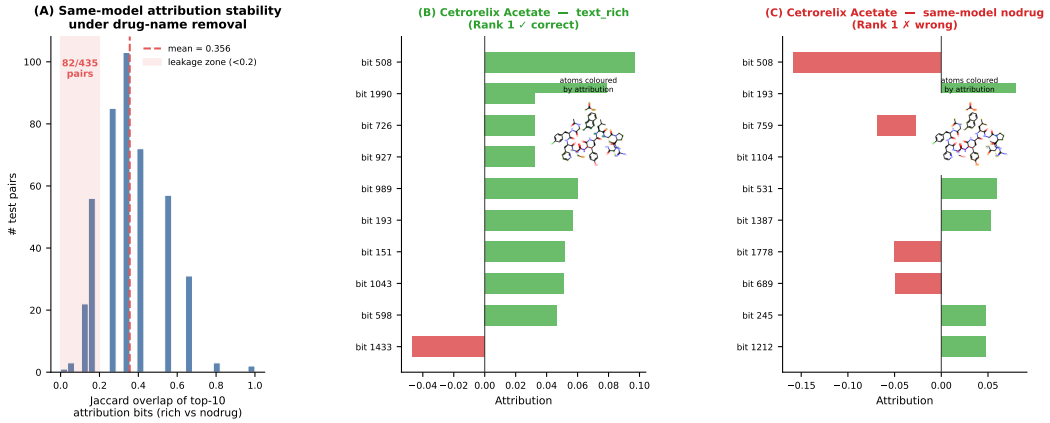

Figure 9: **Same-model drug-name ablation shifts salient substructures.** The distribution of Jaccard overlaps shows that 82 test pairs fall below $J < 0.2$; the Cetrorelix acetate example illustrates how saliency changes when the explicit name is removed.

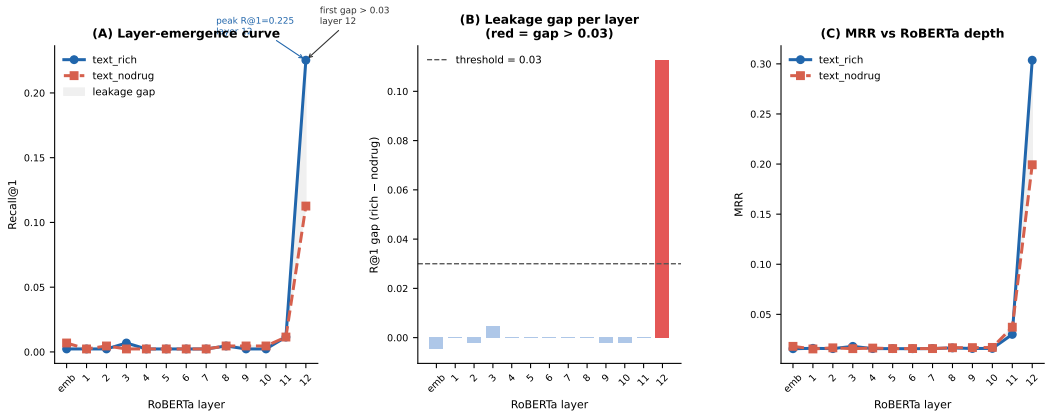

Figure 10: **Contrastive logit lens across RoBERTa layers.** Retrieval is near chance until the final transformer layer, where both rich retrieval signal and the no-name gap emerge.

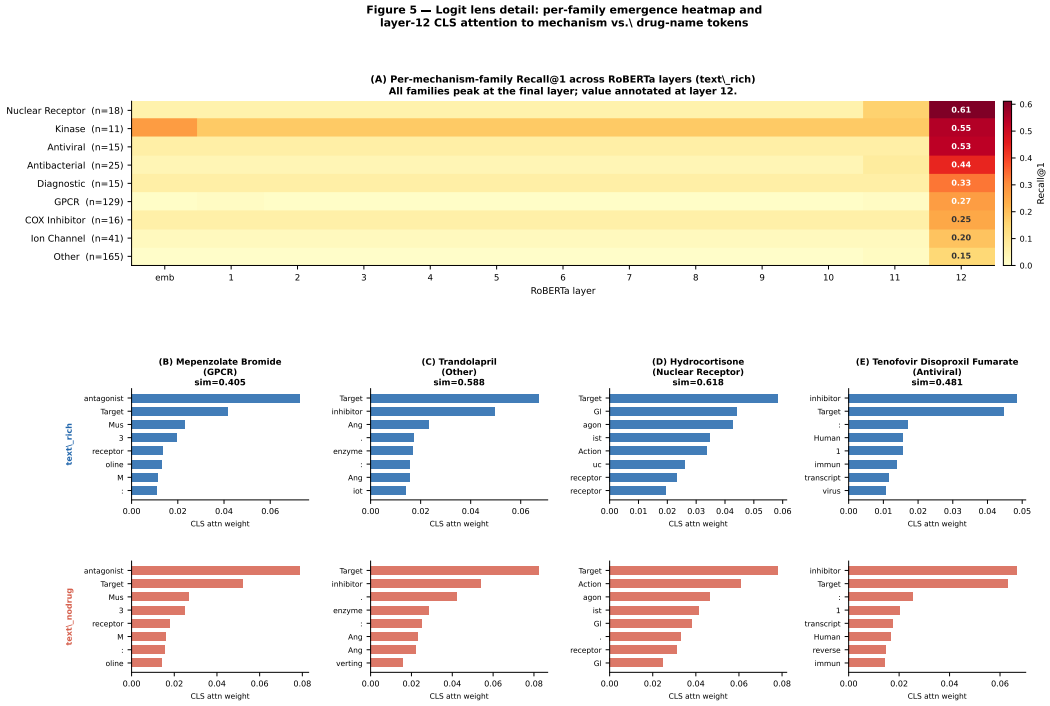

Figure 11: **Logit-lens detail.** Family-level layer-12 retrieval and CLS attention examples under rich and no-drug text conditions.